

Smooth optimization

Let us start simple and assume, f is a smooth function, with L -Lipschitz gradients

$$\|\nabla f(w_1) - \nabla f(w_2)\| \leq L \|w_1 - w_2\|.$$

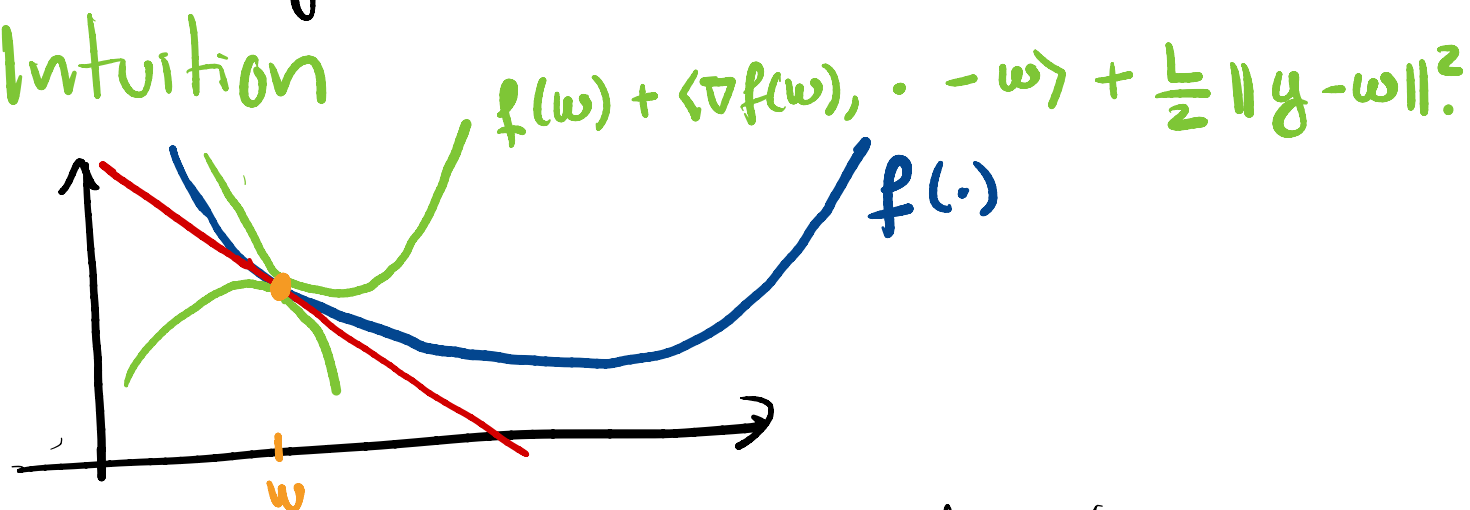
Let's think of f as a black box (we don't know its exact structure). We can merely query $w \mapsto (f(w), \nabla f(w))$. Our goal is to find a ϵ -critical point.

Theorem (Taylor): Let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be an L -smooth function, then

Exercise $|f(y) - f(w) - \langle \nabla f(w), y - w \rangle| \leq \frac{L}{2} \|y - w\|^2 \rightarrow$

Armed with this result we can derive an incredibly useful lemma.

Intuition



This simple result, motivates an incredibly powerful idea.

Algorithmic design principle

Instead of minimizing f directly, iteratively minimize an approximation of f centered at the previous iterate.

We will use this principle over and over again. Let's see a first instantiation.

Step $k \geq 0$:

$$w_{k+1} \leftarrow \operatorname{argmin}_w \left\{ \underbrace{\langle \nabla f(w_k), w - w_k \rangle}_{\text{Linear approximation}} + \underbrace{\frac{1}{2\eta_k} \|w - w_k\|^2}_{\text{Anchor}} \right\}.$$

$\Phi_k()$

Let's use first order optimality conditions.

$$0 = \nabla \Phi(w_{k+1}) = \nabla f(w_k) + \frac{1}{\eta_k} (w_{k+1} - w_k)$$

$\Downarrow$

(GD) $w_{k+1} = w_k - \eta_k \nabla f(w_k).$

Next, let's try to show this is a good idea.

Descent lemma: Let f be L -smooth

and $\eta \in (0, 2/L)$. Then

$$f(w - \eta \nabla f(w)) \leq f(w) - \underbrace{\eta \left(1 - \frac{L\eta}{2}\right)}_{\frac{1}{2L} \text{ when } \eta = \frac{1}{L}} \|\nabla f(w)\|^2.$$

Proof: This follows from Taylors with $y = w - \eta \nabla f(w)$, since

$$\begin{aligned} f(y) - f(w) &+ \eta \|\nabla f(w)\|^2 \\ &= f(y) - f(w) + \langle \nabla f(w), y - w \rangle \\ &\leq \frac{L}{2} \|y - w\|^2 \\ &= \frac{L\eta^2}{2} \|\nabla f(w)\|^2. \end{aligned}$$

□

Theorem (@): Suppose that f is L -smooth and $\inf f > -\infty$. Consider the iterates

$$w_{k+1} \leftarrow w_k - \frac{1}{L} \nabla f(w_k).$$

Then,

$$\min_{\substack{k \in [T-1] \\ k=0,1,\dots,T-1}} \|\nabla f(w_k)\| \leq \sqrt{\frac{2L(f(w_0) - \inf f)}{T}}.$$

Proof: Applying the descent lemma to each index and adding all inequalities

ties up yields

$$\begin{aligned}\frac{1}{2L} \sum_{k=0}^{T-1} \|\nabla f(w_k)\| &\leq \sum_{k=0}^{T-1} f(w_k) - f(w_{k+1}) \\ &= f(w_0) - f(w_T) \\ &\leq f(w_0) - \inf f.\end{aligned}$$

Using that the minimum is smaller than the average gives

$$\begin{aligned}\min_{k \in [T-1]} \|\nabla f(w_k)\|^2 &\leq \frac{1}{2L} \sum_{k=0}^{T-1} \|\nabla f(w_k)\|^2 \\ &\leq f(w_0) - \inf f.\end{aligned}$$

Compleating the proof. $\square$

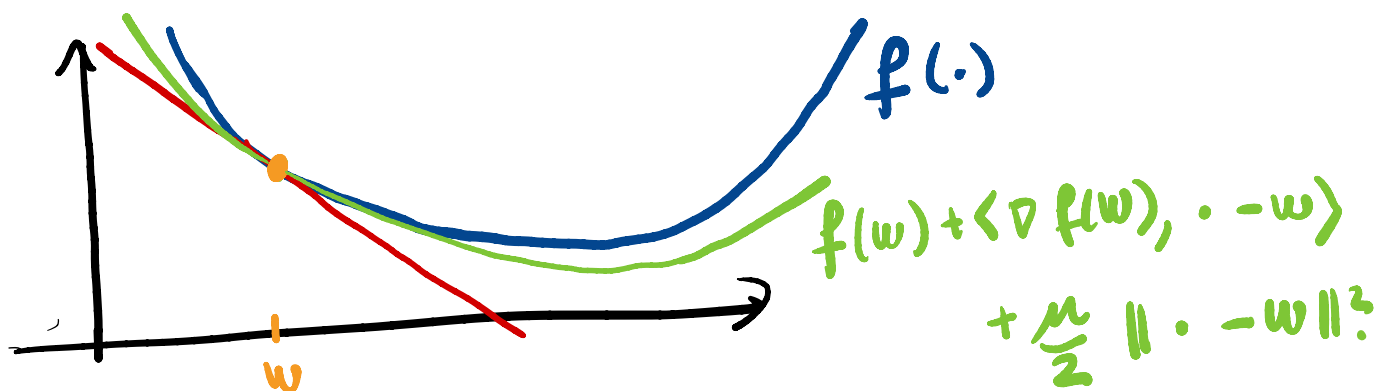
A natural next question is when is it that we can guarantee something about the distance to a solution. For that we need additional regularity assumptions.

Definition: We say that function f is μ -strongly convex if for all $w, y \in \mathbb{R}^d$,

$$(v) \quad f(y) \geq f(w) + \langle \nabla f(w), y - w \rangle + \frac{\mu}{2} \|y - w\|^2.$$

convex when $\mu=0$. $\rightarrow$

Intuition



There are very strong consequences of this assumption.

Proposition (Quadratic growth): Suppose that f is μ -strongly convex. Then, f has a unique minimizer w^* , and for all $w \in \mathbb{R}^d$

$$\frac{\mu}{2} \|w - w^*\|^2 \leq f(w) - f(w^*).$$

Theorem (13): Suppose that f is L -smooth and μ -strongly convex. Then, the iterates of (GD) satisfy

$$f(w_k) - f(w^*) \leq \left(1 - \frac{\mu}{L}\right)^k (f(w_0) - f(w^*)).$$

A simple computation yields that if we want to reach

$$(\downarrow) \quad f(w_T) - f(w^*) \leq \varepsilon,$$

then it suffices to take

(why?)
$$T \geq \frac{L}{\mu} \ln \left(\frac{f(w_0) - f(w^*)}{\epsilon} \right).$$

Further quadratic growth ensures

$$\|w_T - w^*\|^2 \leq 2\epsilon/\mu.$$

Remark: For Theorem (C) we needed

$$T \geq \frac{2L}{\epsilon^2} (f(w_0) - f(w^*)) \text{ for (D)}. \text{ If } \epsilon = 10^{-6}$$

$$\Rightarrow \epsilon^{-2} = 10^{12} \text{ while } \ln\left(\frac{1}{\epsilon}\right) \approx 14. \quad +$$

Thus, how fast (GD) converges is dictated by the ratio

$$\kappa = L/\mu \in [1, \infty].$$

From now on we will call this scalar the **condition number**. This is not an artifact of the proof. (see demo).

We will come back to the proof of Theorem (E), but summarizing:

- ▶ For general smooth functions, we can merely find critical points at a rather slow rate $O(\epsilon^{-2})$.

▷ If we want to guarantee ^{fast} convergence towards a minimizer, we need additional geometric structure.

Polyak-Łojasiewicz Inequality.

For a long time, most of the "additional geometric structure" optimizers used to prove guarantees amounted to strong convexity.
 I'll use this word to refer to people
 In the last couple of decades, that has changed.
 Published this in '63 and was ignored until 2016

Definition (PL Inequality): A differentiable function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies the PL-inequality if there is a $\mu > 0$ s.t.

$$\mu (f(x) - \inf f) \leq \frac{1}{2} \|\nabla f(x)\|^2 \quad \forall x \in \mathbb{R}^d. \quad \rightarrow$$

This is a relatively strong assumption, in particular it yields that

$$\nabla f(x) = 0 \quad \Rightarrow \quad f(x) = \inf f.$$

Nonetheless, it generalizes the notion of strong convexity.

Lemma (SC $\Rightarrow$ PL). Suppose $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is

differentiable and μ -strongly convex, then it satisfies μ -PL. Further, there exists a nonconvex, smooth function that satisfies μ -PL.

Proof: We only prove the first statement. Pick $w^* \in \arg\min f$ and set $y = w^*$ in (V) to obtain

$$f(w^*) \geq f(w) + \langle \nabla f(w), w^* - w \rangle + \frac{\mu}{2} \|w - w^*\|^2$$

Complete the square $\Rightarrow$

$$f(w) + \frac{\mu}{2} \left[\left\| \frac{\nabla f(w)}{\mu} - (w^* - w) \right\|^2 - \frac{1}{\mu^2} \|\nabla f(w)\|^2 \right]$$

$$\geq f(w) - \frac{1}{2\mu} \|\nabla f(w)\|^2.$$

Rearranging the terms gives the result. $\square$

Now, let's state and prove the linear convergence of (GD) under this condition.

Theorem (13.43): Suppose that f is L -smooth and satisfies μ -PL. Then, the iterates of (GD) with $\eta_k = 1/L$ satisfy

$$f(w_k) - \inf f \leq (1 - \frac{\mu}{L})^k (f(w_0) - f(w^*)).$$

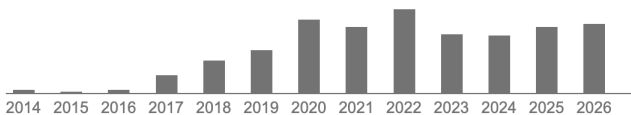
Proof: Using the descent lemma and μ -PL, yields

$$\begin{aligned} f(w_k) - \underbrace{\inf f}_{f^*} &\stackrel{\text{Descent}}{\leq} f(w_{k-1}) - f^* - \frac{1}{2L} \|\nabla f(w_{k-1})\|^2 \\ &\stackrel{\text{PL}}{\leq} f(w_{k-1}) - f^* - \frac{\mu}{L} (f(w_{k-1}) - f^*) \\ &= (1 - \frac{\mu}{L}) (f(w_{k-1}) - f^*) \\ &\stackrel{\text{Recurse}}{\leq} (1 - \frac{\mu}{L})^k (f(w_0) - f^*). \quad \square \end{aligned}$$

History trivia

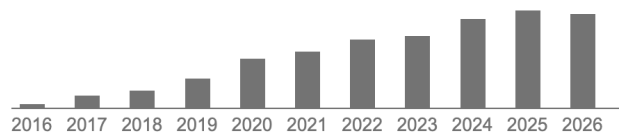
Polyak's paper in '63

Cited by 569



Karimi, Nutini, Schmidt 2016

Cited by 2041



Next class we will see modern applications of this result.

Conclusions

Slide from a 2021 talk by Polyak (Passed away in 2023)

- Many classes of optimization problems and many methods were invented "purely theoretically" without orientation on real-life problems. However later practical applications appear where the methods happen to be highly effective and well fitted for.
- But to wait you have to live long